Mathematical Models in Biotechnology

Jeffrey D. Varner
Robert Frederick Smith School of Chemical and Biomolecular Engineering,
Cornell University, Ithaca, NY, USA

Abstract

This chapter develops flux balance analysis as a linear-programming framework for predicting metabolic phenotypes from stoichiometry alone.

1 Introduction

Biotechnology depends on predicting and directing cellular metabolism. Metabolic engineering redirects flux toward a target product, bioprocess development tunes feeding and environment to sustain it, and strain design asks which genetic changes will move a phenotype in a wanted direction; each is, at bottom, a question about the intracellular flux distribution, the rates at which the reactions of metabolism actually run under a given condition [1]. That distribution is what a quantitative model is asked to predict, and it is also what is hardest to observe. A genome sequence and the stoichiometry of its annotated reactions are knowable in full, but the internal fluxes, the kinetic parameters that would set them, and the metabolite concentrations that drive them are, at genome scale, largely unmeasured. The problem is therefore to predict a metabolic phenotype from what can be known rather than from what a direct calculation would require.

Two broad strategies answer that problem. Mechanistic kinetic models write an explicit rate law for every reaction, from Michaelis–Menten saturation [2] through mass action to the power-law formalism of biochemical systems theory [3, 4], and integrate the resulting equations forward in time; they resolve the dynamics in full but demand a rate constant, an affinity, or a kinetic order for every step, parameters that are rarely available beyond a handful of well-studied pathways. Constraint-based models make the opposite trade. They ask only for the stoichiometry of the network and impose conservation of mass at steady state, and in doing so they scale to genome-wide reconstructions of thousands of reactions [1, 5]. Flux balance analysis is the constraint-based workhorse: it casts phenotype prediction as a linear program over the flux vector [6, 7]. Conservation alone, however, does not determine a flux distribution; it fixes a whole subspace of them, and what selects a prediction from that subspace is the box of upper and lower bounds placed on each reaction. The bounds, not the stoichiometry, are where biological information enters, and the question this chapter takes up is how it enters: through which bound does a thermodynamic estimate, a turnover number, an expression measurement, or a regulatory signal actually change a flux prediction?

This chapter develops flux balance analysis as that integrative framework, organized around the bounds as the gateway through which every class of biological information enters a flux estimate. The flux constraint itself is derived not by asserting steady state but from an open mole balance on a growing culture, a derivation that keeps the growth-dilution term the textbook form discards and writes the constraint as $\mathbf{S}\hat{\mathbf{v}} = \mu\mathbf{x}$ rather than $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$. A single bound is then read as a product of factors, one gateway each for thermodynamic reversibility, enzyme kinetics, gene expression, and regulation, so that opening a gateway means supplying the data or the model that sets its factor. Two worked steady-state examples make the construction concrete: a urea-cycle metabolic network whose flux distribution is pinned by thermodynamic and kinetic data drawn entirely from public databases, and a feedback-inhibited pathway whose regulatory truth, generated independently from a kinetic model, is recovered by opening the regulatory gateway on a single bound. Two published systems then show the same gateways opened together at a scale no hand calculation could reach, one estimating fluxes in a dynamic cell-free reaction [8] and one designing a glycan-producing strain by model-guided gene deletion [9]. Read across these cases, estimating a flux distribution from data and designing one by intervention are the same operation on the bounds, run in opposite directions.

2 From Mole Balances to the Flux Constraint

Flux balance analysis (FBA) rests on an algebraic constraint rather than a differential equation, and that constraint is a mole balance in disguise. Writing the balance out, instead of asserting metabolic steady state by fiat, makes explicit what is assumed away when the constraint is reduced to its familiar textbook form, and which operating conditions the reduction cannot cover. Consider a well-mixed bioreactor of volume V holding a growing cell culture. The culture exchanges mass with its surroundings through a set of physical streams s (feed, harvest, gas sparging) and hosts an intracellular reaction network with n fluxes $\hat{v}_j$, $j = 1, \ldots, n$, acting on m metabolite species. For a species i present at concentration C_i, an open mole balance over the whole culture accounts for material entering and leaving through the streams together with material produced or consumed by the reaction network:

$$\sum_s d_s\, C_{i,s}\, \dot{V}_s \;+\; \sum_j \sigma_{ij}\hat{v}_j\, V \;=\; \frac{d}{dt}\big(C_i V\big), \tag{1}$$

where $d_s = +1$ for streams entering the reactor and $d_s = -1$ for streams leaving it, $C_{i,s}$ is the concentration of species i in stream s, $\dot{V}_s$ is the volumetric flow rate of that stream, and σ_{ij} is the stoichiometric coefficient of species i in reaction j, the (i, j) entry of the stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{m\times n}$. Equation (1) is exact: it presumes nothing about growth rate, feeding strategy, or steady state, and it is the starting point for the reduction carried out below.

Two steps turn Equation (1) into a statement about fluxes alone. First, write the reactor volume in specific units, $V = B\bar{V}$, with B the total biomass in the reactor and $\bar{V}$ the culture volume per unit biomass; this ties the extensive balance to the biomass that is actually growing, and the specific growth rate follows as $\mu \equiv (1/B)\, dB/dt$. Second, expand the accumulation term by the product rule,

$$\frac{d}{dt}\big(C_i V\big) = \frac{d}{dt}\big(C_i B\bar{V}\big) = B\bar{V}\,\frac{dC_i}{dt} + C_i\bar{V}\,\frac{dB}{dt} + C_i B\,\frac{d\bar{V}}{dt}. \tag{2}$$

Dividing Equation (1) through by $V = B\bar{V}$ and substituting Equation (2) isolates the intensive balance for species i,

$$\frac{1}{B\bar{V}}\sum_s d_s\, C_{i,s}\, \dot{V}_s \;+\; \sum_j \sigma_{ij}\hat{v}_j \;=\; \frac{dC_i}{dt} \;+\; \mu\, C_i \;+\; \frac{C_i}{\bar{V}}\,\frac{d\bar{V}}{dt}, \tag{3}$$

where the middle term on the right, μC_i with $\mu = (1/B)\, dB/dt$, is the growth-dilution term produced by the product rule. Three assumptions each remove one of the remaining terms. If the intracellular pool of species i is at pseudo-steady state ($dC_i/dt = 0$, the usual separation-of-timescales argument, since metabolite turnover on the order of seconds to minutes is fast relative to a doubling time of hours), if the specific culture volume is constant ($d\bar{V}/dt = 0$, so the working volume tracks the growing biomass rather than drifting relative to it), and if species i crosses no physical stream (the sum over s vanishes for this species), Equation (3) collapses to

$$C_i\,\mu \;=\; \sum_j \sigma_{ij}\hat{v}_j \qquad \Longleftrightarrow \qquad \boxed{\mathbf{S}\hat{\mathbf{v}} \;=\; \mu\mathbf{x}}, \tag{4}$$

where $\mathbf{x} = (C_1, \ldots, C_m)^\top$ collects the intracellular concentrations and $\hat{\mathbf{v}} \in \mathbb{R}^n$ collects the fluxes. Equation (4) is the honest flux constraint: net metabolic production of every intracellular species is balanced not against zero, but against the rate at which that species is diluted by its own growing culture.

The form used throughout the constraint-based modeling literature is obtained by discarding the right-hand side of Equation (4) outright:

$$\mathbf{S}\hat{\mathbf{v}} \;=\; \mathbf{0}. \tag{5}$$

Equation (5) is a labeled special case of Equation (4), not an independent starting point [6]. It is a good approximation exactly when the two conditions that justify dropping $\mu\mathbf{x}$ hold together: growth is slow relative to metabolic turnover, and intracellular pools are dilute enough that $\mu\mathbf{x}$ stays small in absolute terms even when μ itself is not negligible. Both conditions can fail in operating regimes of real interest. In fed-batch culture the feed is metered to increase the working volume over the course of a run, so the specific volume drifts ($d\bar{V}/dt \neq 0$); the assumption used to reach Equation (4) no longer holds, and the discarded term carries information about the feed schedule that Equation (5) cannot see. In cell-free systems there is no growing cell to dilute into ($\mu = 0$), yet the reaction mixture is manifestly not at steady state: transcript, protein, and metabolite pools accumulate and decay over the course of a batch reaction ($d\mathbf{x}/dt \neq 0$), so the pseudo-steady-state assumption fails from the other direction. Neither shortfall is a corner case; a cell-free capstone example later in this chapter returns to exactly this setting, where the term Equation (5) discards is not a formal nuisance but a quantity under direct experimental measurement.

Equation (4) and its special case Equation (5) are, as written, statements about a closed system: every species that appears has its production and consumption accounted for entirely by the internal network $\mathbf{S}\hat{\mathbf{v}}$, with the physical stream terms of Equation (1) already discarded along the way. Real metabolism is open, since cells take up substrates and secrete products across the boundary of whatever system is being modeled. Constraint-based reconstructions restore that openness without reintroducing the stream terms explicitly: hypothetical exchange, or boundary, reactions are added to the network itself, one pseudo-reaction $\emptyset \rightleftharpoons M_i$ for each boundary metabolite M_i, entering $\mathbf{S}$ as an isolated nonzero column with no reaction partner [1]. Flux through an exchange reaction enters Equation (4) or Equation (5) exactly as any metabolic flux does, but it is now interpretable as uptake or secretion across the system boundary rather than as a step of intracellular chemistry. A closed algebraic statement can thereby represent a genuinely open system: openness becomes entirely a matter of which reactions are admitted into $\mathbf{S}$ and how their bounds are set, the question taken up next.

3 The Linear Program and Its Geometry

The honest constraint of Equation (4), together with its steady-state special case Equation (5), is a statement of conservation and nothing more: it does not by itself pick out any particular flux distribution among all those consistent with it. Turning conservation into a prediction requires two further ingredients. The first is a box of flux bounds, $\boldsymbol{\ell} \leq \hat{\mathbf{v}} \leq \mathbf{u}$, one lower and one upper bound per reaction, encoding thermodynamic reversibility ($\ell_j < 0$ permitted), enzyme capacity, and substrate availability; how those bounds are set is taken up next. The second is an objective, a linear functional $\mathbf{c}^\top \hat{\mathbf{v}}$ of the flux vector that states what the cell, or the modeler, is presumed to optimize. With both in hand the mole-balance constraint becomes a linear program:

$$\begin{aligned} \max_{\hat{\mathbf{v}} \in \mathbb{R}^n} \quad & \mathbf{c}^\top \hat{\mathbf{v}} \\ \text{subject to} \quad & \mathbf{S}\hat{\mathbf{v}} = \mathbf{0} \quad (\text{or } \mu\mathbf{x}), \\ & \boldsymbol{\ell} \leq \hat{\mathbf{v}} \leq \mathbf{u}. \end{aligned} \tag{6}$$

Equation (6) is flux balance analysis in its complete, textbook form [6]. The objective vector $\mathbf{c}$ is a choice, not a derived quantity: setting $c_j = 1$ at the index of a biomass pseudo-reaction, one that drains precursors in the ratios required to build new cell mass, casts the LP as growth-rate maximization, the default assumption for a cell under selection pressure to divide. Setting $c_j = 1$ instead at the index of a product-export exchange reaction casts the same LP as yield maximization for a chosen metabolite, the framing relevant to a bioprocess engineer optimizing a production strain rather than modeling wild-type physiology. Nothing in Equation (6) distinguishes these cases mechanistically; the constraint set is identical, and only $\mathbf{c}$ changes.

The linear-programming machinery is necessary, rather than a matter of convenience, because the constraint set alone is almost always underdetermined. A genome-scale reconstruction lists every reaction annotated to a genome, and the reaction count n exceeds the metabolite count m by a wide margin, often an order of magnitude or more, because a single metabolite typically participates in several reactions while a single reaction typically touches only a handful of metabolites. The stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{m \times n}$ therefore has many more columns than rows, and its null space $\mathcal{N}(\mathbf{S}) = \{\hat{\mathbf{v}} \in \mathbb{R}^n : \mathbf{S}\hat{\mathbf{v}} = \mathbf{0}\}$ is nontrivial: a whole subspace, not a point, satisfies the steady-state constraint. Every flux vector in $\mathcal{N}(\mathbf{S})$ conserves mass at every internal node exactly as well as every other; conservation alone cannot distinguish among them. The flux bounds cut this unbounded subspace down to a bounded polytope, and the objective in Equation (6) selects a single vertex, or a face, of that polytope, but the underlying indeterminacy, that most of the space allowed by stoichiometry is not resolved by stoichiometry, does not go away merely because an optimum has been found.

The singular value decomposition makes the geometry of that indeterminacy explicit. Writing $\mathbf{S} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$, with $\mathbf{U} \in \mathbb{R}^{m \times m}$ and $\mathbf{V} \in \mathbb{R}^{n \times n}$ orthogonal and $\boldsymbol{\Sigma} \in \mathbb{R}^{m \times n}$ diagonal with nonnegative entries, the columns of $\mathbf{V}$ associated with zero singular values span $\mathcal{N}(\mathbf{S})$, the right null space: every feasible steady-state flux mode is a linear combination of these columns, and reading Equation (6) through this lens, the LP is a search over coefficients of that combination rather than over the raw flux vector. The columns of $\mathbf{U}$ associated with zero singular values span the left null space of $\mathbf{S}$, equivalently the null space of $\mathbf{S}^\top$; a vector $\mathbf{y}$ in this space defines a conserved moiety, a linear combination $\mathbf{y}^\top \mathbf{x}$ of metabolite concentrations, such as a bound cofactor pair sharing a fixed pool total, that is invariant under the reaction network regardless of which flux vector in $\mathcal{N}(\mathbf{S})$ is realized. For a genome-scale reconstruction the rank of $\mathbf{S}$ is smaller than the reaction count, $\text{rank}(\mathbf{S}) \ll n$, so the right null space, and with it the space of flux distributions the constraints alone cannot distinguish, is large [1]. That indeterminacy can be read reaction by reaction. Because an LP optimum need not be unique, and because many reactions can vary freely without changing the objective value at all, flux variability analysis reports a range rather than a point for each flux: with the objective fixed at its optimal value $Z^\star$ (or at a specified fraction of it), each v_j is in turn minimized and maximized subject to Equation (6)'s constraints [7]. A reaction whose minimum and maximum coincide is pinned by the network topology and bounds regardless of objective choice, while a reaction with a wide range is left essentially free, and the pattern of which fluxes fall into which category is exactly the information that a single optimal flux vector, taken alone, cannot report.

4 Bounds as Gateways

Conservation alone leaves the flux distribution underdetermined: the mole-balance constraint fixes a subspace, not a point, and the box of bounds together with the objective is what narrows that subspace to a prediction. The objective is a modeling choice. The bounds are where biology enters, and they enter not as numbers to be looked up but as a model in their own right. Writing a single upper bound as a product of factors makes that model visible. For reaction j,

$$-\delta_j \Big[V^{\circ}_{\max,j}\,(e/e^{\circ})\,\theta_j(\cdot)\,f_j(\cdot)\Big] \;\leq\; \hat{v}_j \;\leq\; V^{\circ}_{\max,j}\,(e/e^{\circ})\,\theta_j(\cdot)\,f_j(\cdot), \tag{7}$$

where $V^{\circ}_{\max,j}$ sets a reference capacity carrying the units of flux and the remaining factors are dimensionless corrections that scale that capacity up or down. Each factor is a gateway for a distinct class of biological information: δ_j admits thermodynamics, $V^{\circ}_{\max,j}$ and f_j admit enzyme kinetics, (e/e°) admits gene expression, and θ_j admits regulation. A bound is therefore an honest ledger of what is known about a reaction. With no information beyond a capacity scale the correction factors are set to unity, and each is reopened, one gateway at a time, as measurements or mechanistic models become available. This is also where the underdetermined polytope is finally narrowed: not by conservation but by the bounds, and every gateway that closes tightens that polytope with information of a particular kind, so that tracing a prediction back to its cause is a matter of asking which factor in Equation (7) was responsible.

The thermodynamic gateway is the switch δ_j, which decides whether the reaction may carry flux in the reverse direction at all. It takes one of two values,

$$\delta_j \;=\; \begin{cases} 1, & \text{reaction } j \text{ reversible,} \\ 0, & \text{reaction } j \text{ irreversible,} \end{cases} \tag{8}$$

so that $\delta_j = 1$ leaves the lower bound of Equation (7) negative and the reaction free to run backward, while $\delta_j = 0$ collapses the lower bound to zero and pins the reaction to the forward direction. The assignment follows the sign of $\Delta G^{\circ}_j - \Delta G^{*}_j$, the standard free energy of reaction measured against a threshold ΔG^{*}_j fixed by the physiological concentration range, or equivalently a cutoff on the equilibrium constant $K_{\mathrm{eq},j}$: a reaction whose driving force cannot plausibly change sign anywhere in that range is declared irreversible, and its reverse flux is forbidden. Standard free energies for this test are drawn from group- and component-contribution estimates, for which eQuilibrator provides genome-scale coverage [10].

When the reaction is allowed to proceed, its capacity is set by enzyme kinetics, and this is the role of the reference scale $V^{\circ}_{\max,j}$ together with the saturation factor f_j. The scale is a maximal velocity, the product of a turnover number and a characteristic enzyme abundance,

$$V^{\circ}_{\max,j} \;=\; k^{\circ}_{\mathrm{cat},j}\,e^{\circ}, \tag{9}$$

where $k^{\circ}_{\mathrm{cat},j}$ is the catalytic rate constant of the enzyme catalyzing reaction j and e° is a reference enzyme concentration. Turnover numbers are tabulated per enzyme and substrate in BRENDA [11], and a representative e° can be read from compilations of cellular abundances such as BioNumbers

[12]. The scale $V^{\circ}_{\max,j}$ is the flux the enzyme would sustain at reference abundance and saturating substrate, and the fraction of that ceiling actually reached at the prevailing substrate concentration is the saturation factor. That factor follows from the enzyme mechanism itself. A single-substrate enzyme binds substrate reversibly and turns it over to product,

$$E + S_j \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} [ES] \xrightarrow{k_{\text{cat}}} E + P, \tag{10}$$

with $[ES]$ the enzyme-substrate complex. Holding the complex at quasi-steady state, on the argument that it equilibrates fast relative to the depletion of substrate,

$$\frac{d[ES]}{dt} = k_1[E][S_j] - (k_{-1} + k_{\text{cat}})[ES] = 0, \tag{11}$$

and eliminating the free enzyme through the conservation of total enzyme $[E]_T = [E] + [ES]$, gives the complex in terms of the substrate alone,

$$[ES] = \frac{[E]_T\,[S_j]}{K_{M,j} + [S_j]}, \qquad K_{M,j} \equiv \frac{k_{-1} + k_{\text{cat}}}{k_1}. \tag{12}$$

The reaction velocity is the rate of the turnover step, $v_j = k_{\text{cat}}[ES]$, so writing the capacity as $V_{\max,j} = k_{\text{cat}}[E]_T$ recovers the Michaelis–Menten rate law [2], and the saturation factor is its normalization to that capacity,

$$v_j \;=\; V_{\max,j}\,\frac{[S_j]}{K_{M,j} + [S_j]} \qquad \Longrightarrow \qquad f_j([S_j]) \;=\; \frac{v_j}{V_{\max,j}} \;=\; \frac{[S_j]}{K_{M,j} + [S_j]} \in [0,1], \tag{13}$$

with $[S_j]$ the substrate concentration and $K_{M,j}$ the Michaelis constant. The saturation factor f_j interpolates between an empty gateway ($f_j \to 0$ as substrate is exhausted) and a fully open one ($f_j \to 1$ at saturation), and it is the same hyperbolic law that governs single-enzyme assays. Because f_j depends on the substrate concentration, opening this gateway makes the bound a function of the metabolic state rather than a fixed number, the point at which a static flux balance begins to acquire the state-dependence that a dynamic treatment carries to its conclusion.

The scale $V^{\circ}_{\max,j}$ was fixed at a reference enzyme abundance e°, but the abundance actually present is itself a variable set by gene expression, and the ratio (e/e°) is the gateway through which that variability enters. To treat abundance as a modeled quantity rather than a fixed parameter, write balances for the transcript m_j and the protein p_j of the gene encoding the enzyme,

$$\dot{m}_j = r_{X,j}\,u_j - (\theta_{m,j} + \mu)\,m_j + \lambda_j, \tag{14}$$

$$\dot{p}_j = r_{L,j}\,w_j - (\theta_{p,j} + \mu)\,p_j, \tag{15}$$

where $r_{X,j}$ and $r_{L,j}$ are transcription and translation rate capacities, u_j and w_j are dimensionless control variables that gate those capacities, $\theta_{m,j}$ and $\theta_{p,j}$ are first-order degradation rate constants for message and protein, and λ_j is a basal, or leak, source of transcript. The loss term in each balance is not the degradation constant alone but the sum $(\theta_{m,j}+\mu)$ or $(\theta_{p,j}+\mu)$: every intracellular species is removed both by its own turnover and by dilution into the growing culture at the specific growth rate μ. That μ is not a new parameter. It is the very growth-dilution term retained in the honest constraint, Equation (4), carried through consistently: an enzyme is diluted by growth exactly as a metabolite is, and the same μ that balances metabolic production in the flux constraint sets the residence time of the transcripts and proteins that fix the flux bounds. Holding the balances at steady state gives closed-form abundances,

$$m_j^{\star} = \frac{r_{X,j}u_j + \lambda_j}{\theta_{m,j} + \mu}, \qquad p_j^{\star} = \frac{r_{L,j}w_j}{\theta_{p,j} + \mu}, \qquad (e/e^{\circ}) = \frac{p_j^{\star}}{e^{\circ}}, \tag{16}$$

and the protein level $p_j^\star$ is mapped onto the abundance of the catalytic activity for reaction j through the gene-protein-reaction (GPR) rules of the reconstruction, which encode whether an activity requires a single gene product, several assembled into a complex, or any one of several interchangeable isozymes. Through Equation (16) the expression gateway couples the flux ceiling to the entire upstream apparatus of transcription and translation, with μ standing in every denominator as the visible signature of the growth-aware bookkeeping begun in the mole-balance derivation. The ratio (e/e°) is also the natural port for measured data: a transcriptomic or proteomic profile enters here, rescaling the capacity of each reaction by the abundance actually observed under a given condition and turning one reconstruction into a family of condition-specific models.

The factor θ_j is a control function, a dimensionless number in $[0, 1]$ reporting the fraction of a reaction's capacity switched on by the regulatory state of the cell, and the same machinery that sets it supplies the control variables u_j, w_j, and the leak λ_j left unspecified in the expression balances. A regulated element, a promoter or an allosteric enzyme, visits a discrete set of microstates s, and the fraction of time spent in each follows a Boltzmann distribution over the state energies,

$$p_s = \frac{f_s\, e^{-\beta\epsilon_s}}{Z}, \qquad Z = \sum_s f_s\, e^{-\beta\epsilon_s}, \tag{17}$$

where ϵ_s is the energy of microstate s, $\beta = 1/(k_B T)$, Z is the partition function that normalizes the distribution, and f_s is a Hill-type function of the effector that stabilizes state s, $f_s = x^n/(K^n + x^n)$ in an activating effector concentration x. Summing the state probabilities over the microstates that permit activity, and splitting that set into states reached through the regulated input $\mathcal{A}$ and states occupied without it $\mathcal{B}$, gives the total occupancy together with its basal part,

$$\bar{u} = \sum_{s\in\mathcal{A}} p_s + \sum_{s\in\mathcal{B}} p_s = u + u^\dagger, \qquad \lambda \equiv r_X\, u^\dagger. \tag{18}$$

The regulated occupancy u is the control that appears in Equation (14), the basal occupancy $u^\dagger$ is exactly the leak source λ of that same balance, and the normalized activity $\bar{u}$ is the control function θ_j entering the bound: the regulatory and expression gateways are two views of one coupled process rather than independent knobs. This partition-function control function, Equations (17) and (18), is not a phenomenological Hill curve grafted onto a rate law. It is the effective biophysical model of Adhikari and coworkers [13], one object in which the Boltzmann weighting over states and the control function it produces are derived from the same statistical-mechanical accounting rather than assembled from two separate parts.

Every gateway just opened demands data that is often unavailable: thermodynamic estimates, turnover numbers, substrate concentrations, expression measurements, and regulatory parameters. When that data is scarce the correction factors are set to their information-free defaults, $f_j \approx 1$, $(e/e^\circ) \approx 1$, and $\theta_j \approx 1$, and the general bound collapses to a two-parameter box,

$$-\delta_j\, V^\circ_{\max,j} \;\le\; \hat{v}_j \;\le\; V^\circ_{\max,j}. \tag{19}$$

Equation (19) retains only the thermodynamic gate and the capacity scale, and it is exactly the reversible or irreversible $\pm V_{\max}$ box used throughout genome-scale flux balance analysis. It is the tractable baseline, the honest statement of a bound when nothing beyond a capacity estimate is known, and Equation (7) records precisely which assumptions that baseline makes and which gateway must be reopened to relax each of them.

The expression and regulatory gateways lumped transcription and translation into aggregate capacities $r_{X,j}$ and $r_{L,j}$, but these processes can themselves be resolved to the level of the sequences they read. Transcription of a gene and translation of its protein consume nucleotides, amino acids, and energy in stoichiometric ratios fixed by the gene and protein sequences, and those precursor demands can be computed directly from the sequences and entered into $\mathbf{S}$ as reactions in their own right [5], with the precursors supplied by exchange reactions across the system boundary. Expression then becomes flux-bearing chemistry inside the same linear program as metabolism, a construction carried out for cell-free protein synthesis by Vilkhovoy and coworkers [14], and it is the mechanism behind the cell-free capstone taken up later in this chapter.

5 Worked Example: Urea-Cycle Metabolism

The general bound of Equation (7) carries four gateways, and a genome-scale reconstruction may in principle open all four at once. The urea cycle in HL-60 cells is small enough to make the opposite case concretely: two gateways, thermodynamic and kinetic, filled entirely from public databases, already pin down a sharp prediction without a single measurement taken on the cells themselves. The network retains the same steady-state constraint $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$, Equation (5), and the same linear program, Equation (6), established earlier. What changes from one application to the next is only the content poured into $\mathbf{S}$, $\boldsymbol{\ell}$, $\mathbf{u}$, and $\mathbf{c}$.

The reconstruction couples the four canonical urea-cycle enzymes to a fifth, competing branch and to the extracellular environment. Argininosuccinate synthetase (v_1, EC 6.3.4.5) condenses citrulline with aspartate and ATP to argininosuccinate, releasing AMP and diphosphate; argininosuccinate lyase (v_2, EC 4.3.2.1) cleaves that intermediate to arginine and fumarate; arginase I (v_3, EC 3.5.3.1) hydrolyzes arginine to ornithine and urea; and ornithine transcarbamylase (v_4, EC 2.1.3.3) recombines ornithine with carbamoyl phosphate to regenerate citrulline, closing the cycle. A fifth reaction, nitric oxide synthase (v_5, EC 1.15.13.39), draws on the same arginine pool consumed by v_3 and oxidizes it with NADPH and oxygen to nitric oxide and citrulline, diverting flux away from urea production toward signaling chemistry. These five enzymatic reactions act on eighteen metabolites. Fourteen of the eighteen, carbamoyl phosphate, aspartate, ATP, water, oxygen, NADPH, and protons on the uptake side, urea, fumarate, AMP, diphosphate, orthophosphate, $NADP^+$, and nitric oxide on the secretion side, cross the system boundary through fourteen exchange reactions, b_1 through b_{14}, written with the import-positive convention $\emptyset \to M_i$ established earlier. The remaining four, arginine, citrulline, ornithine, and argininosuccinate, are pure cycle intermediates with no exchange reaction of their own: each is produced and consumed entirely inside the network. Collecting five enzymatic and fourteen exchange fluxes against eighteen metabolite balances gives $\mathbf{S} \in \mathbb{R}^{18\times 19}$, $m = 18$ intracellular balances and $n = 19$ reactions.

The thermodynamic gateway of Equation (8) was opened first. Standard Gibbs free energies of reaction for all five enzymatic steps were evaluated by eQuilibrator [10]: $\Delta G^\circ_{v_1} = -4.3$, $\Delta G^\circ_{v_2} = -5.5$, $\Delta G^\circ_{v_3} = -51.0$, $\Delta G^\circ_{v_4} = -30.3$, and $\Delta G^\circ_{v_5} = -1220$ kJ/mol. The threshold ΔG^*_j of Equation (8) was fixed here at -10 kJ/mol: a reaction whose standard free energy falls below that cutoff is declared irreversible and its reverse flux forbidden ($\delta_j = 0$), while a reaction whose free energy sits above it is left reversible ($\delta_j = 1$). Only v_1 and v_2 cleared the threshold and remained reversible; v_3, v_4, and v_5 fell below it, v_5 the furthest at -1220 kJ/mol, and were pinned irreversible. Every exchange reaction was left unconstrained by this test and kept a wide default bound in both directions, since the direction of nutrient uptake or product secretion is a matter of medium composition rather than of standard free energy.

The kinetic gateway of Equation (9) was opened next. Turnover numbers for the five enzymes were taken from BRENDA [11] and combined with a common reference enzyme abundance $e^\circ = 0.01$ mmol/gDW, a characteristic order of magnitude for intracellular enzyme concentration drawn from compilations such as BioNumbers [12]. Argininosuccinate synthetase and nitric oxide synthase share $k^{\circ}_{\text{cat}} = 10.0\ \text{s}^{-1}$, giving $V^{\circ}_{\max,v_1} = V^{\circ}_{\max,v_5} = 0.100\ \text{mmol gDW}^{-1}\,\text{h}^{-1}$; arginase I has $k^{\circ}_{\text{cat}} = 190.0\ \text{s}^{-1}$, giving $V^{\circ}_{\max,v_3} = 1.9\ \text{mmol gDW}^{-1}\,\text{h}^{-1}$; and ornithine transcarbamylase has $k^{\circ}_{\text{cat}} = 410.0\ \text{s}^{-1}$, giving $V^{\circ}_{\max,v_4} = 4.1\ \text{mmol gDW}^{-1}\,\text{h}^{-1}$. Argininosuccinate lyase is the outlier: $k^{\circ}_{\text{cat},v_2} = 3.28\ \text{s}^{-1}$, an order of magnitude below the other four turnover numbers, gives $V^{\circ}_{\max,v_2} = 0.0328\ \text{mmol gDW}^{-1}\,\text{h}^{-1}$, the smallest of the five capacities by more than threefold. Every exchange reaction was again left with a wide capacity bound, since a boundary reaction represents bulk transport rather than a single characterized enzyme.

With the thermodynamic and kinetic gateways open and the other two left shut, the general bound of Equation (7) collapses to the two-parameter box of Equation (19) for every enzymatic reaction: no substrate-concentration data enters the model, so the saturation factor is held at its information-free default $f_j = 1$; enzyme abundance is fixed at the reference value itself, so $(e/e^{\circ}) = 1$; and no regulatory information is imposed, so $\theta_j = 1$. The bound used in the linear program is therefore $-\delta_j V^{\circ}_{\max,j} \le \hat{v}_j \le V^{\circ}_{\max,j}$ for v_1 through v_5, with δ_j and $V^{\circ}_{\max,j}$ as assigned above, and $-1000 \le \hat{v}_j \le 1000\ \text{mmol gDW}^{-1}\,\text{h}^{-1}$ for the fourteen exchange reactions, wide enough to be effectively unconstrained. With the feasible box fixed, the one remaining ingredient is the objective, here the maximization of urea export. Exchange reaction b_4 represents urea crossing the system boundary and, following the same import-positive convention, is written $\emptyset \to$ urea, so that positive $\hat{v}_{b_4}$ denotes uptake and negative $\hat{v}_{b_4}$ denotes secretion. Maximizing secretion is therefore equivalent to minimizing $\hat{v}_{b_4}$, and since the linear program of Equation (6) is stated as a maximization, the objective coefficient is set to $c_{b_4} = -1$ and every other entry of $\mathbf{c}$ to zero: maximizing $\mathbf{c}^{\top}\hat{\mathbf{v}} = -\hat{v}_{b_4}$ drives $\hat{v}_{b_4}$ as negative as possible, which is exactly maximal urea export.

Solving Equation (6) by linear programming [6] with these bounds and this objective gave a single, sharply determined answer. The entire enzymatic backbone, v_1 through v_4, saturated at $0.0328\ \text{mmol gDW}^{-1}\,\text{h}^{-1}$, the capacity imposed by argininosuccinate lyase, and the nitric oxide synthase branch v_5 carried zero flux: with arginine supply capped by the same v_2 bottleneck that limits the rest of the cycle, none of it was left over to spend on the competing NOS reaction once urea export is the sole objective. The optimal urea-export magnitude was $0.0328\ \text{mmol gDW}^{-1}\,\text{h}^{-1}$, attained at $\hat{v}_{b_4} = -0.0328$, and every exchange flux carrying carbon or nitrogen into or out of the cycle, carbamoyl phosphate and aspartate on the uptake side, urea and fumarate on the secretion side, scaled to match it, giving the full optimal flux distribution (Fig. 1).

This result was reached without a single concentration, expression level, or regulatory parameter measured in HL-60 cells: every bound came from a public database, eQuilibrator for the sign of δ_j and BRENDA for the scale of $V^{\circ}_{\max,j}$, read against a textbook reference enzyme abundance. Two of the four gateways in Equation (7) were opened, and two were left at their default of unity, and the flux distribution of Figure 1 is the direct consequence: a single hard bottleneck at v_2 and a silent competing branch at v_5, both explainable entirely from the reversibility and capacity assigned above. Reopening the regulatory gateway θ_j, so that a reaction's capacity depends on the cell's own control state rather than sitting fixed at its database-derived ceiling, is the subject of the feedback-inhibition example that follows.

6 Worked Example: A Feedback-Inhibited Linear Pathway

The urea-cycle example opened two of the four gateways in Equation (7), thermodynamic and kinetic, and left the regulatory factor θ_j at its information-free default of unity. This example opens that third gateway, and it does so on a pathway simple enough that the correct answer can be obtained two different ways and checked against itself: once from an explicit kinetic model

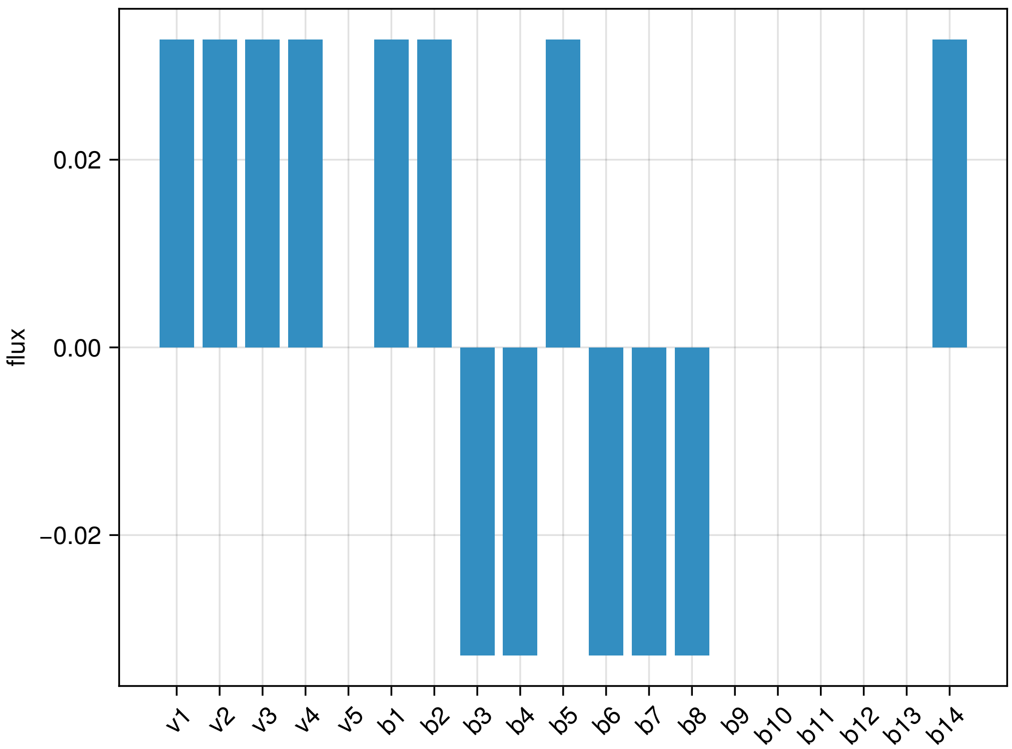

Figure 1: Optimal flux distribution for the urea-cycle linear program of Equation (6), with bounds set by the thermodynamic and kinetic gateways and the objective $c_{b_4} = -1$ maximizing urea export. Bar heights give the flux $\hat{v}_j$ ($\mathrm{mmol\,gDW^{-1}\,h^{-1}}$) at the optimum. The enzymatic backbone v_1 through v_4 saturates at 0.0328, the capacity of argininosuccinate lyase (v_2), while the nitric oxide synthase branch v_5 carries zero flux. Exchange fluxes b_1 through b_{14} follow the import-positive convention, so negative bars denote net secretion; b_4 (urea) shows the maximized export flux of -0.0328 $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$.

integrated to steady state, and once from the linear program of Equation (6), with the kinetic result standing in as the truth the flux-balance calculation must reproduce. The pathway is a linear chain,

$$\emptyset \xrightarrow{r_0} X_1 \xrightarrow{r_1} X_2 \xrightarrow{r_2} X_3 \xrightarrow{r_3} \emptyset, \tag{20}$$

with r_0 the uptake step, r_1 and r_2 interconversions, and r_3 the export step, and the end product X_3 inhibits the input reaction r_0: a high steady-state level of X_3 throttles its own supply. This is textbook end-product inhibition, and it is deliberately the simplest network that can carry it, a single control point acting on a single upstream step.

The truth was generated from a Biochemical Systems Theory (BST) S-system, the power-law formalism in which every reaction rate is written as a product of the participating species raised to real-valued kinetic orders, $v_j = \alpha_j \prod_i X_i^{g_{ij}}$ [3, 4]. Ordinary mass-action kinetics recovers integer orders; feedback inhibition is written just as naturally with a negative order, so that increasing the inhibiting species lowers rather than raises the rate. Here r_0 carries a negative kinetic order in X_3,

$$v_{r_0} = \alpha_{r_0} X_3^{-1/2}, \qquad \alpha_{r_0} = 10, \tag{21}$$

with α_{r_0} the uninhibited capacity of the uptake step and the exponent $-1/2$ the strength of the feedback, while r_1, r_2, and r_3 are ordinary first-order steps with rate constants 100, 100, and 1 respectively, the first two set fast enough that they never limit throughput. The model was built and integrated to steady state with BSTModelKit.jl [15]. Because the chain of Equation (20) is linear, every reaction shares a single steady-state flux, the throughput $T = v_{r_0} = v_{r_1} = v_{r_2} = v_{r_3}$, and because r_3 is first order with unit rate constant, $X_3 = T$ at steady state. Substituting into Equation (21) gives a self-consistency condition on T alone,

$$T = \alpha_{r_0} T^{-1/2} \implies T = \alpha_{r_0}^{2/3} = 10^{2/3} \approx 4.64, \tag{22}$$

the value confirmed by direct numerical integration of the S-system. Feedback has throttled the pathway to a throughput less than half of the uninhibited capacity $\alpha_{r_0} = 10$, and it is precisely that gap, between what the input step could do unregulated and what it actually does once its own product pushes back, that the flux-balance side of this example must recover without ever being told the answer.

The same regulatory fact admits two representations, and making that duality explicit is the point of this example. In the S-system it is the negative kinetic order in Equation (21): the exponent on X_3 in the rate law for r_0. In the flux-balance picture it is the control function θ_j of Equation (7), the same object introduced earlier through the partition-function accounting of Equations (17) and (18) [13]. There θ_j was built from a Boltzmann sum over regulatory microstates; here, for a single allosteric step acting on a single effector, that general machinery collapses to the concrete power law $\theta(X_3) = X_3^{-1/2}$, the same exponent that appears in Equation (21). The two formalisms are not independent descriptions that happen to agree. They are the same biology, end-product inhibition of the uptake step, written once as a kinetic order inside a dynamic rate law and once as a control factor multiplying a static capacity, and the worked comparison below shows that either route, when driven by the same measured X_3, lands on the same throughput.

The flux-balance model retains the stoichiometry of Equation (20) in a matrix $\mathbf{S} \in \mathbb{R}^{3\times4}$, rows X_1, X_2, X_3 and columns $r_0, \ldots, r_3$, and the same steady-state constraint $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$ established earlier and carried into the linear program. The three downstream steps r_1, r_2, and r_3 were left with a generous constant capacity, above the uptake ceiling, so that they never bind and the input reaction remains the sole determinant of throughput, exactly as it is in the S-system. All the regulatory content sits on the single bound that matters,

$$0 \le \hat{v}_{r_0} \le V^{\circ}_{\max,r_0}\, \theta(X_3), \qquad V^{\circ}_{\max,r_0} = 10, \tag{23}$$

the regulatory instance of Equation (7) with the thermodynamic, expression, and saturation gateways left at their defaults and only θ in play. The objective maximizes export, $\hat{v}_{r_3}$, so that the optimal throughput is read directly off the bound on the single reaction that constrains it.

Two experiments were run against Equation (23), differing in nothing but θ. With the regulatory gateway closed, θ was forced to its information-free default of unity and the input bound sat at the full uninhibited capacity, $\hat{v}_{r_0} \leq 10$. Solving the linear program under that bound drove every flux in the chain to 10, the naive optimum overshooting the true throughput of 4.64 by more than a factor of two, because a feedback-blind model has no way to know that the cell itself would never let X_3 climb high enough to sustain that rate. With the regulatory gateway open, θ was instead evaluated at the measured steady-state end-product level, $\theta = X_3^{-1/2} \approx 0.46$, computed from $X_3 \approx 4.64$ through the same control-function machinery that produced Equation (21), not copied from the throughput answer itself. The input bound tightens to $\hat{v}_{r_0} \leq 10 \times 0.464 \approx 4.64$, and the linear program, maximizing the same objective under the same stoichiometry and the same downstream capacities, now returned a throughput of 4.64 across all four reactions, matching the BST truth to within numerical tolerance. Because the two runs differ in exactly one number, the value of θ multiplying the input capacity, and that single change is what separates a threefold overshoot from the correct answer, the regulatory gateway was shown to be load-bearing rather than decorative: turning it off provably changes the prediction, and turning it back on provably restores the truth. The per-reaction fluxes for all three cases, the BST truth, the gateway-closed overshoot, and the gateway-open recovery, appear alongside the uninhibited capacity line at 10 (Fig. 2).

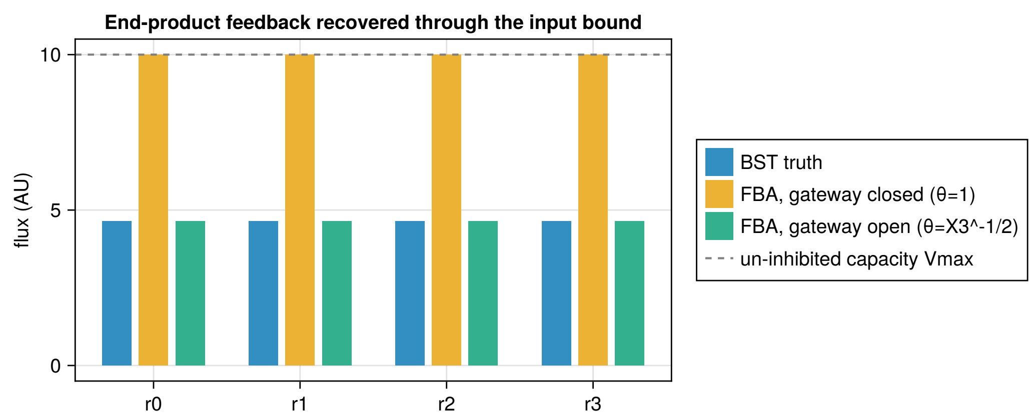

Figure 2: Per-reaction steady-state fluxes for the linear feedback-inhibited pathway of Equation (20). Bars compare the BST S-system truth (throughput $T \approx 4.64$), the flux-balance solution with the regulatory gateway closed ($\theta = 1$, overshooting to the uninhibited capacity of 10 on every reaction), and the flux-balance solution with the gateway open ($\theta = X_3^{-1/2} \approx 0.46$ computed from the measured end-product level, recovering the truth). The dashed line marks the uninhibited capacity $V^{\circ}_{\max,r_0} = 10$.

The mole-balance constraint alone leaves flux underdetermined, a subspace rather than a point, and it is the bound, not the stoichiometric matrix, that carries whatever biology narrows that subspace to a prediction. This example makes both statements concrete for regulation specifically. Nothing about end-product inhibition appears anywhere in $\mathbf{S}$: the stoichiometry of Equation (20) is the same whether or not X_3 feeds back on r_0. The entire regulatory fact, encoded in the S-system as a negative kinetic order and in the flux-balance model as the control function θ on the input bound, is restored through Equation (23) alone, and the two representations, checked against each other on this pathway, agree.

7 Integration in Practice

The urea-cycle and feedback examples opened the gateways of Equation (7) one at a time, on networks small enough that a hand calculation could check the result against a known truth. Two published systems show the same gateways opened together, at a scale no hand calculation could reach, and driven by measurement rather than by a modeler's guess at what a bound should be. The first couples metabolism and regulation into one dynamic model of a cell-free reaction; the second uses the same bounds not to estimate a flux distribution from data but to design one.

Vilkhovoy and coworkers built the metabolic half of that coupling on a sequence-specific reconstruction of *E. coli* cell-free protein synthesis, in which the transcription and translation reactions required to make a target protein from its loaded DNA and mRNA sequence are computed directly from those sequences and entered into the stoichiometric matrix as flux-bearing chemistry alongside metabolism itself [14], the sequence-to-stoichiometry construction described earlier and credited there as the mechanism behind this capstone. Control over that expression machinery, the fraction of promoter and ribosome capacity actually switched on at a given moment, is supplied by the effective biophysical partition-function model of Adhikari and coworkers [13], the same statistical-mechanical control function of Equations (17) and (18), built earlier from a Boltzmann sum over regulatory microstates and worked by hand for a single feedback loop. In this setting the same θ_j machinery is asked to populate every regulated step of a larger network at once, rather than one control point on one pathway. They then closed the loop dynamically [8]: metabolism, expression, and regulation are integrated into a single constraint-based model whose bounds are recomputed as the reaction proceeds, so that the thermodynamic, kinetic, expression, and regulatory gateways are opened together rather than in isolation. This integration is checked against denser data: time-resolved measurement of the intracellular state over the course of a batch reaction, on the order of sixty metabolites tracked through the reaction, together with time courses of mRNA, protein abundance, and enzyme activity. Every worked example earlier in this chapter compared a flux prediction to a single steady-state target; here the comparison runs against a trajectory, resolved in both time and in the identity of the species tracked, at a resolution the small hand-checked networks never had reason to offer.

This density of measurement is what makes cell-free protein synthesis the natural home for the term that ordinary flux balance analysis sets aside. A cell-free reaction has no growing culture to dilute into, so the specific growth rate μ is zero, but the reaction is manifestly not at steady state: transcript, protein, and metabolite pools rise and fall over the course of the batch, so the pseudo-steady-state assumption that collapses Equation (1) to Equation (4) fails from the direction opposite to slow growth. The accumulation that assumption would otherwise discard is exactly what the sixty-odd tracked metabolites record, and because the reaction vessel and the intracellular compartment are one and the same volume, with no membrane separating a sampled extracellular phase from an inferred intracellular one, that accumulation is read directly off the measurement rather than reconstructed indirectly from an uptake or secretion rate. This is the setting pointed to when Equation (5) was first written down: cell-free integration is where the growth-dilution and accumulation term Equation (5) discards is, for once, under direct experimental observation rather than assumed away.

The second capstone runs the same construction in the opposite direction. Wayman and coworkers grafted the N-linked glycosylation pathway of *Campylobacter jejuni*, the source of the designer glycans used to improve the pharmacokinetic and immunogenic properties of therapeutic glycoproteins, onto a genome-scale reconstruction of *E. coli* metabolism, and then searched the combined network for gene deletions predicted to couple glycan output to growth itself, so that a strain could not grow well without also making the glycan [9]. Where the cell-free capstone read the gateways of Equation (7) from measurement to obtain a flux estimate, this campaign set them directly: a reaction knocked out has its bound collapsed to zero, the capacity gateway of Equation (7) closed rather than left open, and each candidate deletion, or combination of deletions, defines a distinct box $[\boldsymbol{\ell}, \mathbf{u}]$ under which the linear program is re-solved for the resulting flux distribution, entirely inside the model and before a single strain is built. Knocking out a reaction is the simplest way to reach into a bound; the same gateway can in principle be narrowed by attenuating an enzyme's expression or widened by over-expressing it, moving the (e/e°) factor of Equation (7) down or up rather than collapsing it to zero. The deletions this campaign screened were sufficient on their own: the best single knockout identified by the model raised glycan production roughly threefold over the unmodified strain once built, an outcome the model predicted by re-solving the same linear program under the modified bound rather than by any change to the stoichiometry $\mathbf{S}$. The bounds are, in this use, not a report of what the cell is doing but a lever for what the cell is made to do.

Read together, the two capstones are the same operation on Equation (7) run in opposite directions. In the cell-free system, measurement narrows the gateways and the flux distribution is estimated; in the glycan strain, the engineer narrows a gateway and the flux distribution is designed. Nothing about the linear program distinguishes the two uses, since both amount to fixing the bounds $\boldsymbol{\ell}$ and $\mathbf{u}$ and solving the same constrained optimization; only the direction in which information moves through the gateway, from data inward for estimation, from intention outward for design, changes. Estimation and design are, at bottom, the same operation on the bounds, run in two directions.

8 Outlook

The worked examples of this chapter populated the gateways of Equation (7) from tabulated sources: standard free energies from group-contribution estimates, turnover numbers from enzyme databases, a control function built by hand from a partition function over a handful of regulatory microstates. None of that is required by the construction itself. A turnover number, a saturation constant, or a control function θ_j is, mathematically, a parameter or a function fit to data, and where time-resolved measurement of the kind reviewed in the capstones is available, these quantities can be learned directly from the data they are meant to explain rather than looked up from a compilation assembled under different conditions, with the tabulated values retained as a prior where dynamic measurement is not.

The same construction, opened here on networks small enough to check by hand, carries without modification to genome-scale reconstructions of thousands of reactions: Equation (7) is applied reaction by reaction regardless of network size. What changes with scale is the weight the bounds are asked to carry. The null space of $\mathbf{S}$ grows with the excess of reactions over metabolites, so that at genome scale the stoichiometric constraint alone leaves an enormous subspace of flux vectors indistinguishable, and it is precisely there that a poorly constrained gateway, a bound left at its information-free default for want of data, does the most damage to a prediction. Closing gateways with data matters most exactly where the null-space argument bites hardest.

The dynamic, data-rich regime realized in the cell-free capstone is the natural home for the growth-dilution and accumulation term retained in Equation (4) and dropped in Equation (5). Cell-free reactions are one setting dense enough in time-resolved measurement to estimate fluxes without discarding that term, and the same opportunity extends wherever intracellular state can be tracked closely enough in time, in principle inside a growing culture as readily as in a reaction tube. The natural direction is therefore not a retreat to the steady-state reduction of Equation (5) once more data arrives, but the opposite: estimating flux directly against Equation (4), with $\mu\mathbf{x}$ carried rather than discarded, wherever the measurement to support it exists.

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